Grade 6 Accelerated
Unit 6
Lesson 1 Homework

Name______Answer Key
Date______________________
Period______________________

Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

	Exponential Form	Expanded Form	Value
1.	$(2.1)^3$	$(2.1)(2.1)(2.1)$	9.261
2.	$(12)^3$	$(12)(12)(12)$	1728
3.	$\left(\frac{2}{5}\right)^4$	$\left(\frac{2}{5}\right)\left(\frac{2}{5}\right)\left(\frac{2}{5}\right)\left(\frac{2}{5}\right)$	$\frac{16}{625}$
4.	$\left(\frac{3}{2}\right)^4$	$\frac{3}{2}\cdot\frac{3}{2}\cdot\frac{3}{2}\cdot\frac{3}{2}$	$\frac{81}{16}$ or $5\frac{1}{16}$

5. What is the value of the expression $(3.7)^2$?

$(3.7)(3.7) = 13.69$

6. Which expression has the larger value, 3^5 or 5^3? Show your work or explain your reasoning.

$3\cdot3\cdot3\cdot3\cdot3 = 243$

$3^5 > 5^3$ because $243 > 125$.

$5\cdot5\cdot5 = 125$

7. Select all expressions that are equivalent to 4^5.

- [x] $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$
- [ ] 20
- [ ] 5^4
- [x] 1024
- [ ] $4 + 4 + 4 + 4 + 4$

8. Select all expressions that are equivalent to $\left(\frac{1}{8}\right)^3$.

- [ ] $\frac{3}{24}$
- [ ] 512
- [x] $\frac{1}{512}$
- [x] $\frac{1}{8} \cdot \frac{1}{8} \cdot \frac{1}{8}$
- [ ] $\frac{3}{8}$

9. What is the value of the expression $(0.06)^3$?

$(0.06)(0.06)(0.06) = 0.000216$

10. What is the value of the expression $(2.7)^2$?

$(2.7)(2.7) = 7.29$